EV POWERTRAIN
Cooling & Lubrication System

Systems Engineering Reference

Electric Motor · Inverter · Oil Pump · Thermal Management · Drive Cycles · Derating Calibration · Oil Lifetime

Jaguar Land Rover — Electric Drive Unit Programme
Powertrain Thermal Systems Engineering

May 2026

TABLE OF CONTENTS

Chapter 1 — System Overview — EV Cooling Architecture

1.1 Two Primary Thermal Loops
1.2 Oil Circuit (Lubrication & Direct Cooling)
1.3 Coolant Circuit (Indirect Cooling)

Chapter 2 — Oil Pump Sizing — Flow Rate & Pump Speed

2.1 Core Relationships & Heat Rejection Requirements
2.2 Speed-vs-Flow Decision Map
2.3 Pump Speed Scheduling & Lookup Tables

Chapter 3 — Coolant Flow Rate

3.1 Sizing Logic & Key Equations
3.2 Typical Flow Targets by Component
3.3 Coolant Flow Control Strategy

Chapter 4 — Electric Motor Thermal Model (LPTN)

4.1 Lumped Parameter Thermal Network Theory
4.2 Stator Thermal Model Detail
4.3 Rotor Thermal Model Detail

Chapter 5 — Continuous vs Peak Performance

5.1 The Power-Duration Curve
5.2 Temperature Limits by Component
5.3 Thermal Derating Algorithm

Chapter 6 — Drive Cycles & Testing

6.1 Standard Drive Cycles Overview
6.2 Motor Component Tests
6.3 EDU Integration Tests
6.4 Vehicle Level Validation

Chapter 7 — VSC Minimum Oil Flow Request

7.1 What Is a VSC Request
7.2 Reasons for Minimum Flow at Zero Torque/Speed
7.3 Implementation in Software

8.1 Vehicle → EDU → Motor Requirements
8.2 Thermal Budget Allocation
8.3 Flow Requirements Derivation
8.4 V-Model & Verification Closure

9.1 Component Bench Tests
9.2 Motor Component Tests
9.3 EDU Integration Tests
9.4 Vehicle Level Validation
9.5 Requirements Traceability Matrix

10.1 LPTN Structure & ECU Implementation
10.2 Loss Map Generation
10.3 Steady State R Calibration
10.4 Transient C Calibration

10.5 Model Validation Criteria

Chapter 11 — Derating Calibration

11.1 Derate Map Structure
11.2 Setting Threshold Temperatures
11.3 Multi-Dimensional Derating Arbitration
11.4 Rate Limiting & Hysteresis
11.5 Full Derate Validation Test

Chapter 12 — Oil Lifetime Profile

12.1 EV vs ICE Oil Degradation
12.2 Degradation Mechanisms
12.3 Oil Lifetime Profile Construction
12.4 Arrhenius Lifetime Calculation
12.5 Accelerated Aging Tests
12.6 Oil Condition Monitoring

Chapter 13 — Stator Winding Temperature

13.1 Insulation System & Classes
13.2 Thermal Ageing — Montsinger Rule
13.3 Winding Temp & Motor Performance
13.4 Measurement Methods in Development

Chapter 14 — Rotor & Magnet Temperature

14.1 Why Rotor Is the Harder Thermal Problem
14.2 Permanent Magnet Properties vs Temperature
14.3 Demagnetisation — The Irreversible Failure
14.4 Rotor Cooling Strategies
14.5 Stator-Rotor Thermal Coupling

Chapter 15 — Quick Reference Tables

CHAPTER 1

System Overview

EV Cooling & Lubrication Architecture

1.1 The Two Primary Thermal Loops

An EV powertrain thermal management system operates two fundamentally different cooling circuits that must work in concert to protect the motor, inverter, and gearbox. Understanding how these loops interact is the starting point for all thermal engineering work.

Oil Circuit — Lubrication & Direct Cooling

The oil circuit uses Automatic Transmission Fluid (ATF) or a dedicated e-motor oil. This fluid makes direct physical contact with the motor's rotating components and is therefore capable of removing heat from surfaces that coolant can never reach. Key functions include end-winding spray cooling, rotor shaft feed, bearing lubrication, gear mesh lubrication, and differential lubrication.

- Fluid: ATF or dedicated EV e-motor oil (e.g., 0W-20 synthetic)
- Direct contact with copper windings, rotor surfaces, bearings, and gear teeth
- Oil-to-water heat exchanger transfers heat to the coolant circuit
- Typical flow range: 2–10 L/min depending on demand
- Sump temperature limit: 125–130°C for sealed-for-life design

Coolant Circuit — Indirect Cooling

The coolant circuit uses a water-glycol mixture (50/50 by volume) and is completely isolated from rotating and live electrical components. It removes heat from the stator housing jacket and the inverter cold plate, then rejects that heat to the vehicle radiator or chiller via the Vehicle Thermal Management System (VTMS).

- Fluid: Water-Glycol 50/50, density ~1060 kg/m³, Cp ~3600 J/kg·K
- Cools: Stator housing jacket, inverter cold plate, HV battery (separate loop in some architectures)
- Never contacts rotating or electrically live components
- Typical flow range: 10–25 L/min
- Maximum coolant inlet temperature: 70°C at worst case ambient

Parameter	Oil Circuit	Coolant Circuit
Fluid	ATF / EV Oil (0W-20)	Water-Glycol 50/50
Density	~870 kg/m³	~1060 kg/m³
Specific Heat (Cp)	~2000 J/kg·K	~3600 J/kg·K
Typical Flow	2–10 L/min	10–25 L/min
Temp Range	–40°C to 130°C	–40°C to 115°C
Pump Type	Electric Oil Pump (EOP)	PWM Electric Water Pump
Heat Sink	Oil-to-Water HEX → Coolant	Vehicle Radiator / Chiller
Key Interfaces	Windings, bearings, gears	Stator housing, cold plate

CHAPTER 2

Oil Pump Sizing

Flow Rate & Pump Speed Determination

2.1 Core Relationship — Heat Rejection Drives Flow

A fundamental principle in EV thermal engineering is that oil flow demand is driven primarily by the heat rejection requirement, not by motor speed alone. This is in contrast to an ICE transmission where oil flow is often mechanically linked to engine speed. The electric oil pump (EOP) in an EV is fully decoupled, allowing independent thermal and lubrication control across all operating conditions.

```
Q_required = P_loss / (rho x Cp x delta_T_allowable) Q = volumetric flow rate (m3/s or L/min) P_loss = heat generated in component (W) rho = oil density (~870 kg/m3 for ATF) Cp = specific heat capacity (~2000 J/kg.K) delta_T = allowable temperature rise across component (K)
```

2.2 Speed-vs-Flow Decision Map

Operating Condition	Speed Range	Primary Heat Source	Oil Flow Need
Low speed / high torque (launch, hill start)	0–2000 RPM	High I²R copper losses, bearing load	HIGH — maximum flow required
Mid-range cruise	3000–6000 RPM	Mixed copper + iron losses	Medium flow
High speed, light load	8000–15000 RPM	Windage losses, churning dominant	Moderate (churning losses rise sharply)
Peak / launch event	Any speed	Maximum total losses everywhere	MAXIMUM — pump at ceiling speed
Stationary with torque (hill hold)	0 RPM	Sustained I²R, zero windage	HIGH — no centrifugal oil distribution aid

2.3 Pump Speed Scheduling

Most modern EV EDUs use an electric oil pump (EOP) that is fully software-controlled. The pump speed is determined by a multidimensional lookup table arbitrated by the thermal control strategy. The final pump speed command is the maximum of all contributing demand sources.

Pump Speed Arbitration Logic (Pseudocode)

```
EOP Target RPM = MAX( lookup_2D(motor_torque, motor_speed), // performance demand lookup_1D(max(T_winding, T_oil_sump)), // thermal demand VSC_minimum_flow_request // system floor ) // Convert RPM to actual flow via pump curve: actual_flow = pump_curve(EOP_target_RPM, system_back_pressure)
```

The pump H-Q curve (pressure head vs volumetric flow at each RPM) is the critical calibration input. Without an accurate H-Q curve, all flow estimates in the ECU are wrong. This curve must be measured on a flow bench at multiple oil temperatures because viscosity strongly affects pump delivery.

CHAPTER 3

Coolant Flow Rate

Sizing, Targets & Control Strategy

3.1 Coolant Sizing Logic

The coolant circuit must remove the combined heat load from the inverter cold plate and the stator housing jacket. The fundamental sizing equation is the same as for oil but uses water-glycol fluid properties. The temperature rise across the circuit (delta-T) is a key design choice — smaller delta-T means better cooling but requires higher flow rate and therefore more pump power.

```
Q_coolant = (P_inverter_loss + P_stator_jacket_loss) / (rho_wg x Cp_wg x delta_T)
Water-Glycol 50/50 properties: rho = 1060 kg/m3 Cp = 3600 J/kg.K Typical delta_T
across cold plate: 5 to 15 degC
```

Component	Typical Flow (L/min)	Max Inlet Temp	Typical delta-T
Inverter cold plate	8–15	65°C	5–10°C
Stator housing jacket	10–20	70°C	8–15°C
Combined EDU coolant loop	15–25	70°C	10–15°C

3.2 Coolant Flow Control Strategy

- Low load: Minimum flow to reduce pump parasitic power loss and maximise range
- High load or hot ambient: Maximum pump speed — full flow to maintain component temperatures
- Thermal soak (post-drive): Run-on pumping continues after ignition off to pull residual heat from inverter and motor housing before shutdown
- VSC coordinates with HVAC system — if AC compressor is running, radiator inlet temperature is managed jointly
- PWM water pump: Typically controlled 0–100% duty cycle via thermal controller CAN message

CHAPTER 4

Electric Motor Thermal Model

Lumped Parameter Thermal Network (LPTN)

4.1 LPTN Theory

The motor thermal model is the most important analytical tool in EV cooling engineering. It models the motor as a network of thermal resistances (R) and thermal capacitances (C) — an exact analogy to an electrical RC circuit where heat flow replaces current and temperature replaces voltage. This analogy is powerful because circuit analysis techniques can be applied directly to thermal problems.

```
Heat Flow (W) = Temperature Difference (K) / Thermal Resistance (K/W) [analogous to I = V/R] Thermal Capacitance (J/K) = Mass (kg) x Specific Heat Cp (J/kg.K) [stores heat energy, causes thermal lag] Time Constant: tau = R x C (seconds)
```

Key Thermal Nodes in the Motor Model

Node	Physical Location	Primary Heat Source	Typical Time Constar
T_copper (hot spot)	Deepest winding in slot	I²R copper loss	30–60 s
T_end_winding	Winding overhang outside slot	I²R copper loss	30–90 s
T_stator_tooth	Iron core between slots	Iron eddy + hysteresis loss	60–180 s
T_stator_yoke	Outer ring of stator iron	Iron loss (lower)	120–300 s
T_housing	Stator outer cylindrical surface	Conducted from stator	300–600 s
T_magnet	Embedded in rotor	Magnet eddy currents, conducted	100–300 s
T_rotor_iron	Rotor lamination stack	Rotor iron losses	100–400 s
T_shaft	Motor shaft	Conducted from rotor	200–500 s
T_bearing	Bearing inner/outer race	Conducted + friction loss	150–400 s

4.2 Stator Thermal Model

The stator is the primary heat source in an IPM motor at most operating conditions. The slot liner — a thin sheet of electrical insulation between the copper and the stator iron — is the dominant thermal resistance in the stator heat path. Its thermal conductivity is typically only 0.2 W/m·K (compared to copper at 400 W/m·K and silicon steel at 25 W/m·K). This is why end-winding oil spray is so thermally effective — it bypasses this slot liner resistance entirely.

Stator Loss Equations

```
Stator heat generation: Copper Loss = I_rms^2 x R_dc x K_ac x (1 + alpha x delta_T) K_ac = AC resistance factor (skin effect, proximity effect) alpha_copper = 0.00393 per degC Iron Loss = k_h x f x B_peak^2 + k_e x f^2 x B_peak^2 k_h = hysteresis coefficient (material-dependent) k_e = eddy current coefficient f = electrical frequency = (RPM/60) x pole_pairs B = peak flux density (Tesla)
```

4.3 Rotor Thermal Model

The rotor presents a fundamentally harder thermal problem than the stator. It rotates at up to 20,000 RPM, making direct coolant access impossible. Heat must conduct outward through solid material — from magnet to rotor iron to shaft to bearing to housing — a thermal path with significantly higher resistance than the stator path. The rotor thermal time constant is therefore 3–5 times longer than the stator, meaning rotor temperature continues rising even after the electrical load is removed. This thermal soak behaviour is critical for post-drive thermal management strategy.

Rotor Heat Source	Speed Dependence	Torque Dependence	Peak Risk Condition
Magnet eddy currents	Strong ($\propto f^2$)	Moderate (via current harmonics)	High speed, field weakening
Rotor iron eddy loss	Strong ($\propto f^2$)	Low	High speed, light torque cruise
Rotor iron hysteresis	Linear ($\propto f$)	Low	High speed operation
Conducted from stator via air gap	Moderate (convection ↑ with speed)	Strong (stator hotter at high I)	Low speed, high torque

CHAPTER 5

Continuous vs Peak Performance

Thermal Constraints & Ratings

5.1 The Power-Duration Curve

Every electric motor has two distinct power ratings: a peak rating valid for a limited duration (typically 10–30 seconds), and a continuous rating that can be sustained indefinitely. The boundary between these regions is entirely defined by the thermal management system. A better cooling system directly translates into a higher continuous rating and a longer peak duration.

```
Continuous Power: Operating point where dT_winding/dt = 0 (thermal equilibrium reached with T_winding < T_limit) Cooling system MUST reject all losses at steady state. Peak Duration: t_peak = (m x Cp x delta_T_allowed) / P_loss_excess m = thermal mass of motor (kg) Cp = weighted specific heat (J/kg.K) delta_T = T_limit - T_initial P_loss = heat generation at peak operating point (W)
```

5.2 Temperature Limits by Component

Component	Continuous Limit	Peak Limit	Failure Mode if Exceeded
Stator winding (Class H)	180°C	200–210°C	Insulation breakdown → short circuit
Stator winding (Class C)	200°C+	220°C+	Insulation carbonisation
Rotor magnet (NdFeB N45SH)	120°C	140–150°C	Irreversible demagnetisation
Bearing (all types)	120°C	150°C	Fatigue spalling, seizure
Oil sump	125°C	135°C	Oil oxidation, varnish, seal failure
IGBT junction (Si)	150°C	165°C	Thermal runaway, device failure
SiC MOSFET junction	175°C	185°C	Bond wire failure, short
Gear steel surface	120°C	140°C	Scuffing, micropitting

5.3 Thermal Derating Algorithm Overview

When component temperatures approach their limits, the ECU activates a controlled torque reduction called thermal derating. The derating strategy must balance hardware protection against driver experience — overly aggressive derating creates customer complaints while insufficient derating risks permanent hardware damage.

Derating Arbitration Logic

```
// Multi-input torque limit arbitration in motor controller: Torque_limit = MIN( f(T_winding), // winding insulation protection f(T_magnet), // demagnetisation protection f(T_junction), // inverter IGBT/SiC protection f(I_phase_limit), // current limit (temp-dependent) f(V_dc, RPM), // voltage/flux limit f(T_bearing), // bearing protection f(T_oil_sump) // oil degradation protection )
```

CHAPTER 6

Drive Cycles & Testing

From Component to Vehicle Level Validation

6.1 Standard Drive Cycles

Cycle	Region	Max Speed	Character	Thermal Significance
WLTP Class 3	Global/EU	131 km/h	Mixed urban + extra-urban + highway, 23 min	Gold standard — homologation requirement
US06	USA	129 km/h	Aggressive, rapid acceleration	Worst-case peak thermal stress
NEDC	Legacy EU	120 km/h	Gentle, low average speed	Too mild — being phased out
RTS-95	EU	120 km/h	Highway-focused, sustained high speed	Sustained thermal load on rotor
FTP-75	USA	91 km/h	Urban, stop-start	Low-moderate thermal load
Custom OEM	OEM-specific	Variable	Grade climbs, towing, trailer	Worst-case use-case validation

6.2 Motor Component Level Tests

Thermal Mapping Test

The motor thermal mapping test is the foundational dataset for the entire thermal engineering programme. It measures steady-state temperatures at every point in the motor's operating envelope and is used to calibrate the LPTN thermal model.

Thermal Mapping Procedure

```
Test Setup: Motor on dyno with controlled external coolant and oil supplies Thermocouples: end winding (x4), stator tooth (x2), yoke (x2), housing (x4), oil in/out, coolant in/out, bearings (x2) Flow meters on both oil and coolant circuits Test Matrix: Torque: 50, 100, 150, 200, 250, 300 Nm Speed: 500, 1000, 2000, 4000, 8000, 12000 RPM T_coolant_in: 40, 65 degC T_oil_in: 60, 90 degC Oil flow: 3, 6, 9 L/min Coolant flow: 10, 15, 20 L/min Procedure per point: 1. Start from cold soak (all temps = 25 degC) 2. Apply constant torque and speed 3. Wait for equilibrium: dT/dt < 0.5 degC/min on all sensors 4. Record all temperatures, flows, pressures 5. Cool down fully before next point Total points: 200-400 steady-state operating points Typical duration: 3-6 weeks of dyno time
```

Continuous Rating Validation

This test proves the motor can sustain its rated continuous torque indefinitely without any temperature exceeding its limit. This is a direct verification of the continuous power requirement cascaded from the vehicle level.

Continuous Rating Test

```
Procedure: 1. Pre-condition motor to warm but stable state 2. Apply continuous torque (e.g., 160 Nm) at worst thermal speed (typically base speed — highest current for given torque) 3. Run until ALL temperatures reach equilibrium (30-90 minutes) Pass Criteria (worst case: T_coolant_in=70 degC, T_oil_in=90 degC): T_winding_steady_state < 175 degC (5 degC margin to 180 degC limit) T_magnet_steady_state < 115 degC (5 degC margin to 120 degC limit) T_bearing_steady_state < 115 degC T_oil_out < 125 degC
```

Peak Rating & Derating Validation

```
Peak Duration Test
Phase 1 — Cold Peak: Motor at 25 degC cold soak Apply peak torque instantly Record time until T_winding hits derate threshold Compare to thermal model prediction Phase 2 — Hot Peak (worst case for customer): Pre-condition at continuous rating until thermal equilibrium Apply peak torque from hot starting condition Record time to derate — will be significantly shorter than cold peak Phase 3 — Derate Behaviour Validation: Deliberately run into derate zone Confirm torque reduction follows calibrated slope Confirm smooth recovery — no hunting or oscillation Measure recovery time after load removal
```

6.3 EDU Integration Level Tests

At EDU level, motor, inverter, and gearbox are tested as an integrated assembly. New thermal phenomena emerge at this level that do not exist in motor-only tests: inverter switching losses heat the coolant before it enters the motor; gearbox churning losses heat the oil before it reaches the end windings; real oil distribution across multiple jets can be measured.

Drive Cycle Simulation on EDU Dyno

```
EDU Drive Cycle Test
Test Objective: Validate complete thermal system against transient drive cycle Setup: EDU on dyno, speed/torque time trace replayed from drive cycle data All real-world components: real inverter, real gearbox, real pump Instrumented with full thermocouple harness Key difference from steady state: TRANSIENT behaviour validated — thermal capacitance C values tested Model must predict temperature TRAJECTORY, not just final value Pass Criteria: T_winding predicted vs measured: within +/-8 degC throughout cycle Temperature PEAKS correctly predicted (drives derating decisions) No temperature limit exceeded at any point during cycle
```

Worst-Case Thermal Stress Tests

Test Name	Procedure	What It Validates
Repeated Launch	Full peak torque 0→100 km/h, coast back, repeat until derate. Count launches.	Peak thermal capacity, derating activation
Sustained Grade Climb	Simulate 10% grade at 120 km/h for 30 minutes continuously	Continuous rating at worst sustained condition
Hot Ambient Grade	As above but with coolant inlet forced to 70°C	True worst-case combined condition
Hot Soak Restart	Motor at 80°C thermal soak, immediately apply peak torque	Derating on hot restart — customer worst case
Cold Start High Load	–20°C cold soak, full torque demand immediately	Oil viscosity, pump priming, cold start protection

6.4 Vehicle Level Validation

Climatic Chamber Testing

Climate Condition	Chamber Setting	Key Tests	Pass Criteria
Hot Climate	+45°C, solar load	WLTP + grade climb + repeated launch	No derating in WLTP; grade without limit breac
Cold Climate	–20°C to –40°C	Cold start, warm-up strategy, first drive	Pump primes within 5s; full torque within 2 min
Altitude Simulation	Reduced pressure	Sustained load at altitude	Thermal margin still adequate
Humidity/Condensation	High RH cycling	Thermal cycling cold-hot	No water ingress to oil; seal integrity

Proving Ground Testing

- Arizona (Death Valley, July): 45–50°C ambient — real-world radiator inlet temperature at maximum; sustained motorway cruise, towing in hills, repeated hard acceleration from traffic lights
- Sweden / Northern Canada (January): –30 to –40°C — cold start oil viscosity validation, pump priming, warm-up time to full performance, heater demand interaction
- Alps / Pikes Peak: Sustained grade climb — worst-case continuous thermal scenario; repeated ascent (back-to-back) for thermal soak validation
- Dubai / Abu Dhabi: 45°C + high humidity — dual stress on cooling and sealing; relevant for Middle East market

CHAPTER 7

VSC Minimum Oil Flow Request

System-Level Flow Floor & Post-Drive Strategy

7.1 What Is a VSC Request?

The Vehicle System Controller (VSC) — sometimes called the VCU (Vehicle Control Unit) — is the top-level vehicle brain that coordinates all subsystems. It sends flow requests to the thermal management system, including the electric oil pump, based on system-level needs that go beyond what the motor torque and speed demand would require. The VSC minimum flow request acts as a hard floor below which the oil pump speed never falls, regardless of the torque/speed map output.

7.2 Reasons for Minimum Flow at Zero Operating Point

Reason	Explanation	Consequence If Not Met
Bearing lubrication	Bearings need minimum oil film even at st[illegible] rotation	[illegible]
Seal conditioning	Lip seals and O-rings require wetted cont[illegible]	[illegible] leakage path formation
Thermal soak protection	After hard driving, motor is thermally hot. [illegible]	[illegible]
Gear pre-lubrication	Before first torque application, gear and d[illegible]	[illegible] prevent scuffing
Thermal model boundary	Thermal observer in ECU needs stable oi[illegible]	[illegible] at rest
Post-drive run-on	Driver switches off; pump continues run[illegible]	[illegible]

7.3 Typical VSC Minimum Flow Values

Operating Condition	Min Oil Flow	Approx Pump Speed	Duration
Ignition ON, cold, stationary	0.5–1.5 L/min	200–500 RPM	Until first drive
Hot soak (T_winding > 80°C), vehicle stopped	1.5–3.0 L/min	500–1000 RPM	Until T_winding < 60°C
Post-drive run-on (ignition off)	1.0–2.0 L/min	400–800 RPM	2–10 minutes
Active driving, zero torque coasting	1.0–2.0 L/min	400–800 RPM	Continuous while moving
Remote pre-conditioning (parked)	1.0 L/min	350–600 RPM	Duration of preconditioning
Full performance demand	6–10 L/min	2000–4000 RPM	Duration of demand

7.4 Software Implementation

EOP Speed Arbitration — ECU Software

```
// EOP speed request arbitration (runs every 10 ms in thermal controller) float torque_speed_req = lookup_2D(motor_torque_Nm, motor_speed_RPM); float thermal_req = lookup_1D(MAX(T_winding_est, T_oil_sump)); float vsc_min_req = CAN_receive(VSC_MIN_FLOW_RPM_signal); // Final command = highest of all requests (MAX wins) float eop_target_rpm = MAX(torque_speed_req, thermal_req, vsc_min_req); // Apply rate limit (prevents pump hunting) eop_target_rpm = rate_limit(eop_target_rpm, prev_rpm, MAX_RAMP_UP, MAX_RAMP_DOWN); // Convert RPM to flow using pump curve interpolation float actual_flow = interp_2D(pump_curve_table, eop_target_rpm, system_back_pressure_bar);
```

CHAPTER 8

Requirements Cascade

Vehicle Level → EDU Level → Motor & Cooling Requirements

8.1 The Cascade Process

Requirements cascade is the systematic process of translating high-level customer and regulatory needs into precise engineering specifications for each component. It is the most important systems engineering activity because every flow rate, temperature limit, and pump speed ultimately exists to fulfil a customer-facing vehicle requirement. Without traceability, a component engineer cannot know whether their specification is correct or whether a design change at component level will violate a vehicle-level promise.

Example Cascade — 0 to 100 km/h Acceleration

```
Acceleration Requirement Cascade
VEHICLE REQUIREMENT: 0-100 km/h in 5.5 seconds Step 1 — Average acceleration: a_avg = (100/3.6) / 5.5 = 5.05 m/s2 Step 2 — Peak tractive force at launch (M=2200 kg): F = 2200 x 5.05 + F_aero + F_rolling = ~12,000 N Step 3 — Wheel torque (R_wheel = 0.35 m): T_wheel = 12,000 x 0.35 = 4,200 Nm Step 4 — EDU output torque (final drive 9.5:1, eta=0.97): T_EDU = 4200 / (9.5 x 0.97) = 456 Nm << EDU REQUIREMENT Step 5 — EDU output speed at 100 km/h: omega_wheel = (100/3.6) / 0.35 = 79.4 rad/s omega_EDU = 79.4 x 9.5 = 754 rad/s = 7,200 RPM << EDU REQUIREMENT
```

8.2 EDU Level Requirements

EDU Requirement	Value	Derived From
Peak output torque	456 Nm	0–100 km/h in 5.5 s
Peak output power	~250 kW	Top speed + grade target
Continuous output torque	180 Nm	20% grade at 80 km/h sustained
Maximum output speed	8,500 RPM	200 km/h top speed
Peak duration	≥ 30 seconds	Product requirement — launch feel
EDU efficiency (WLTP average)	≥ 93%	500 km range target
Coolant inlet temperature max	70°C	System thermal budget
Oil sump temperature max	130°C	Sealed-for-life oil requirement
No derating in WLTP	Full cycle, any ambient to 40°C	Homologation requirement

8.3 Motor Level Requirements

Motor Requirement	Value	Derived From EDU Req
Peak torque	350 Nm (≤ 10 s)	EDU peak torque ÷ gear ratio

Motor Requirement	Value	Derived From EDU Req
Continuous torque	160 Nm (steady state)	EDU continuous ÷ gear ratio
Peak power	230 kW	EDU peak power – inverter losses
Maximum speed	16,000 RPM	EDU max speed × gear ratio
Base speed (corner point)	3,500 RPM	Motor design — MTPA/FW boundary
Peak phase current	800 A_peak	Motor sizing from torque requirement
Continuous phase current	400 A_rms	Thermal model → continuous torque
DC bus voltage	400 V or 800 V	Platform architecture decision
Winding temperature limit	180°C continuous / 200°C peak	Insulation class selection
Magnet temperature limit	120°C continuous / 140°C peak	Magnet grade N45SH properties
Coolant inlet temp max	70°C	EDU level requirement
Oil inlet temp max	90°C	Sump + HEX thermal budget
Minimum coolant flow	10 L/min	Heat rejection capacity calc
Minimum oil flow	3 L/min	End winding temp at continuous
Peak motor efficiency	≥ 96%	EDU efficiency budget

8.4 Thermal Budget Allocation

The total heat rejection budget for the EDU is the product of maximum input power and the complement of EDU efficiency. This budget must be allocated across all loss sources, and the cooling system sized to reject each allocation.

Thermal Budget & Flow Requirement Derivation

```
Total heat rejection budget: P_loss_total = P_input x (1 - eta_EDU) = 250 kW x (1 - 0.93) = 17,500 W Loss allocation: Motor copper loss ~6,000 W (34%) Motor iron loss ~2,500 W (14%) Inverter switching loss ~3,500 W (20%) Inverter conduction loss ~2,000 W (11%) Gearbox mesh loss ~2,000 W (11%) Bearing + windage loss ~1,500 W (9%) ------- Total 17,500 W Coolant circuit rejection (inverter + stator jacket): P_coolant = 3,500 + 2,000 + 6,000 x 0.4 = ~8,000 W Q_coolant = 8,000 / (1060 x 3600 x 10) = 12.6 L/min → spec: 13 L/min Oil circuit rejection (end winding + rotor + gear + bearing): P_oil = 6,000 x 0.6 + 2,500 + 2,000 + 1,500 = ~9,600 W Q_oil = 9,600 / (870 x 2000 x 15) = 3.7 L/min → spec: 4 L/min minimum
```

CHAPTER 9

V-Model Verification & Test Planning

From Component Bench to Vehicle Validation

9.1 The V-Model Structure

The V-model is the governing framework for systems engineering verification and validation. Requirements are decomposed down the left side of the V, and verification evidence closes back up the right side. Every requirement written during design must have a corresponding verification method and test result. The model ensures that component-level test results can be traced all the way back to the original vehicle-level customer promise.

V-Model Structure

```
LEFT SIDE (Requirements Decomposition) Vehicle Requirements | EDU Requirements | Motor / Inverter Requirements | Cooling System Specification | Component Specification (pump, HEX) RIGHT SIDE (Verification Closure — mirror image) Component Bench Test <- validates component spec Motor Dyno Test <- validates motor requirements EDU Integration Test <- validates EDU requirements Vehicle Level Test <- validates vehicle requirements
```

9.2 Component Bench Tests

Oil Pump H-Q Curve Test

Oil Pump Verification Test

```
Objective: Validate pressure-flow characteristic at each RPM and temperature Setup: Pump on calibrated flow bench Flow meter (Coriolis or electromagnetic) Pressure transducers: inlet and outlet Oil bath at controlled temperature: 40, 80, 100 degC Test matrix: RPM: 500, 1000, 1500, 2000, 2500, 3000 Back pressure: 0.5, 1.0, 1.5, 2.0, 3.0 bar Temperature: 40, 80, 100 degC Pass criteria: Flow at 2000 RPM, 1.5 bar >= 6 L/min (per requirement) Volumetric efficiency >= 85% at nominal operating point Flow degradation < 5% after 200,000 km equivalent duty
```

Heat Exchanger Effectiveness Test

HEX Effectiveness Test

```
Key metric: Thermal effectiveness epsilon epsilon = Q_actual / Q_maximum Q_max = C_min x (T_hot_in - T_cold_in) C_min = MIN(m_oil x Cp_oil, m_coolant x Cp_coolant) Pass criteria: epsilon >= 0.75 at nominal flow conditions Oil-side pressure drop <= 0.3 bar at max flow Coolant-side pressure drop <= 0.2 bar at max flow No degradation after 1000-hour thermal cycling test
```

9.3 Motor Component Tests Summary

Test ID	Test Name	Key Measurement	Pass Criterion
MT-2A	Thermal Mapping	T_winding, T_magnet, T_housing at 20 steady-state points	LPTN model error < ±5°C
MT-2B	Continuous Rating	T_winding at thermal equilibrium, continuous torque	T_winding_ss < 175°C at 70°C coolant in
MT-2C	Peak Duration	Time from cold/hot to derate activation	Cold peak ≥ 30 s; hot peak ≥ 10 s

Test ID	Test Name	Key Measurement	Pass Criterion
MT-2D	Oil Flow Sensitivity	T_winding vs oil flow rate	Min flow requirement established with margin
MT-2E	Derating Validation	Torque vs T_winding trajectory	No temperature overshoot beyond T_derate_end
MT-2F	Loss Mapping	P_input – P_output at every operating point	Loss map accuracy ±2% for ECU observer input

9.4 Requirements Traceability Matrix

Requirement	Target Value	Test ID	Test Level	Status
Continuous torque	160 Nm, T<180°C	MT-2B	Motor	PASS
Peak torque duration	350 Nm ≥ 30 s	MT-2C	Motor	PASS
Minimum oil flow	3 L/min at continuous	MT-2D / ET-3A	Motor / EDU	PASS
No WLTP derating	Full 23-min cycle	ET-3B / VT-4A	EDU / Vehicle	PASS
Hot ambient grade	45°C, no derate	VT-4B / VT-4C	Vehicle / PG	ONGOING
Cold start priming	<5 s at –20°C	VT-4B	Vehicle	PLANNED
Oil lifetime	Sealed for life 200k km	OAT-001 / Fleet	Lab / Fleet	ONGOING

CHAPTER 10

Thermal Model Calibration

LPTN Parameter Estimation from Test Data

10.1 The ECU Thermal Observer

The motor thermal model runs inside the ECU in real time at a loop rate of 10–50 ms. In production vehicles, there are no thermocouples in the windings — the ECU model is the only source of winding temperature information. This makes calibration accuracy safety-critical: if the model underestimates winding temperature by 10°C, derating activates too late and insulation damage accumulates. If it overestimates by 10°C, performance is unnecessarily restricted.

ECU Observer Architecture

```
ECU THERMAL OBSERVER Measurable inputs (available in production): I_phase_rms (from inverter current sensors) Motor_speed_RPM (from resolver / encoder) T_coolant_in (from coolant NTC sensor) T_oil_in (from oil NTC sensor) V_dc_bus (from HV system) Estimated outputs (not directly measured in production): T_copper_hotspot -> input to winding derate logic T_magnet -> input to magnet derate / current limit T_bearing -> input to bearing protection logic Model update rate: every 10-50 ms Model complexity: 6-10 thermal nodes (balance accuracy vs CPU load)
```

10.2 Loss Map Generation — Critical First Step

Before any thermal calibration can begin, an accurate loss map must be measured. The loss map defines how much heat is generated at every operating point — it is the heat source input to the LPTN. An error in the loss map creates a systematic error in all thermal predictions regardless of how well the R and C values are tuned.

Loss Map Measurement

```
Loss measurement at each (torque, speed) point: P_loss = P_electrical_in - P_mechanical_out = (V_dc x I_dc) - (Torque x omega) Loss segregation: P_copper = I_rms^2 x R_dc x K_ac x (1 + alpha x delta_T) P_iron = P_loss_total - P_copper - P_mechanical P_mech = P_loss measured at zero current (friction + windage) Temperature correction (CRITICAL): R_copper at 180 degC = R_20 x (1 + 0.00393 x 160) = 1.63 x R_20 Copper losses at hot conditions are 63% higher than cold! Loss map must be 3D: f(torque, speed, temperature)
```

10.3 Steady State R Calibration

R Calibration Optimisation

```
Using steady-state thermal map test data to solve for R values: R4 (housing to coolant): R4 = (T_housing_measured - T_coolant_in) / P_stator_total R_total (copper to coolant, combined): R_total = (T_winding_measured - T_coolant_in) / P_copper R6 (end winding to oil — flow dependent): Calibrate as 2D table: R6 = f(Q_oil, T_oil) At each oil flow test point: R6 = (T_endwinding - T_oil_in) / P_endwinding Optimisation approach: Minimise: SUM_i (T_predicted_i - T_measured_i)^2 Optimise: [R1, R2, R3, R4, R5, R6, R7, R8] Constraints: R > 0, within +/-50% of theoretical Tools: MATLAB fmincon, Python scipy.optimize.minimize
```

10.4 Transient C Calibration

R values control steady-state accuracy. C values control transient accuracy — how fast temperatures rise after a load step. C calibration uses step-load transient test data and requires matching the temperature trajectory, not just the final value.

Transient C Calibration

```
Transient test procedure: 1. Motor at thermal equilibrium at light load 2. Instant step change to high torque 3. Record T_winding every 1 second 4. Tune C values until model matches measured trajectory Physical values (starting point before tuning): C_copper = m_copper x Cp_copper (Cp = 385 J/kg.K — well known) C_stator = m_iron x Cp_iron (Cp_iron = 460 J/kg.K) + m_insulation x Cp_insulation C_housing = m_housing x Cp_aluminium (Cp_al = 900 J/kg.K) Common calibration finding: Effective C_stator < theoretical due to imperfect thermal contact between laminations — tune by 10-30% reduction
```

10.5 Model Validation Acceptance Criteria

Validation Metric	Acceptance Criterion	Why This Matters
T_winding steady state accuracy	±5°C vs measured	Directly sets derate threshold margin
T_magnet steady state accuracy	±8°C vs measured	Demagnetisation risk assessment
T_housing steady state accuracy	±3°C vs measured	Housing is a reference node — errors propagate
Transient trajectory error	±8°C at any time point	Peak prediction accuracy — drives derate activation timing
Drive cycle RMS error (WLTP)	<4°C over full cycle	System-level model validity
Drive cycle peak error	<±5°C at any peak	Most critical — peak drives hardware limits

CHAPTER 11

Derating Calibration

Setting Thresholds, Slopes, Rate Limits & Hysteresis

11.1 Why Derating Calibration Is Critical

Derating calibration is the process of setting the precise temperature thresholds, torque reduction slopes, rate limits, and hysteresis values that govern how the motor controller reduces output when thermal limits are approached. It is a safety-critical calibration because an error in either direction has serious consequences.

Calibration Error Direction	Consequence
Too aggressive (derate too early)	Performance unnecessarily restricted; customer complaint 'car feels slow when warm'; competito
Too conservative (derate too late)	Winding temperature exceeds insulation class limit; accelerated aging; possible catastrophic fail

11.2 Two-Zone Derate Map

Derate Map Structure

```
Torque limit as function of estimated winding temperature: T_winding < T_derate_start:
Torque_limit = 100% (no restriction) T_derate_start <= T <= T_derate_end: Linear reduction
T_winding > T_derate_end: Torque_limit = 0% (full derate) Torque_limit(%) = 100 x (T_derate_end
- T_winding) / (T_derate_end - T_derate_start) Typical values (Class H motor): T_derate_start =
160 degC T_derate_end = 175 degC Recovery temp = 145 degC (hysteresis — see Section 11.4)
```

11.3 Setting T_derate_end

T_derate_end Calibration

```
T_derate_end = T_insulation_class - Safety_margin Insulation classes: Class F: 155 degC rated
continuous Class H: 180 degC rated continuous Class C: >180 degC rated continuous Safety margin
sources: Model prediction error: +/-5 to 8 degC -> margin needed Sensor/input uncertainty: +/-2
to 3 degC -> margin needed Aging factor (end of life): insulation degrades -> extra margin
Regulatory safety factor: programme-specific Typical result for Class H motor: T_derate_end =
170 to 175 degC (180 degC rated minus 5-10 degC model uncertainty margin) Magnet derate end
(separate limit): T_demag (N45SH at zero current) = 180 degC T_demag (N45SH at peak FW current)
= 120 degC T_derate_end_magnet = 130 to 140 degC
```

11.4 Setting T_derate_start

T_derate_start Simulation Method

```
T_derate_start is set by simulation — not by static analysis alone. Simulation approach:
Scenario: Motor at T_derate_start, full peak torque applied Question: Does temperature exceed
T_derate_end before torque reaches zero? Calculate temperature rise rate at peak torque: dT/dt
= P_loss_peak / C_thermal_effective Example: P_loss at peak torque = 8,000 W C_effective
(copper + stator iron) = 15,000 J/degC dT/dt = 8000 / 15000 = 0.53 degC/s Time available before
T_derate_end is reached: t_available = (T_derate_end - T_derate_start) / (dT/dt) = (175 - 160)
/ 0.53 = 28 seconds With rate-limited derate (Nm/s limit), verify torque reaches zero before T
exceeds T_derate_end in simulation. If not -> lower T_derate_start.
```

11.5 Multi-Dimensional Derating Arbitration

Full Arbitration Logic

```
Final torque limit = MIN of all contributing limits: Torque_limit_winding = f(T_winding_estimated) // insulation Torque_limit_magnet = f(T_magnet_estimated) // demag Current_limit_FW = f(T_magnet, RPM) // FW demag Torque_limit_inverter = f(T_junction_estimated) // IGBT/SiC Torque_limit_current = f(I_phase_limit(temp)) // current Torque_limit_voltage = f(V_dc, RPM) // flux Torque_limit_bearing = f(T_bearing_estimated) // bearing Torque_limit_oil = f(T_oil_sump) // oil FINAL = MIN(all_above)
```

11.6 Rate Limiting & Hysteresis

Rate limiting prevents the sudden torque steps that would cause dangerous vehicle lurching. Hysteresis prevents hunting (oscillation in and out of derate). Both must be co-calibrated to ensure the hardware is protected without compromising the stability of the derate control strategy.

Rate Limiting & Hysteresis Calibration

```
Rate Limiting: Maximum derate rate: -50 Nm/s (negative = reducing) From 350 Nm to 0 Nm: minimum 7 seconds Maximum recovery rate: +20 Nm/s (gentler than derate) Verification: dT/dt x t_ramp_down < delta_T_available Example: 0.53 degC/s x 7s = 3.7 degC rise during ramp 3.7 < (175 - 160) = 15 degC -> safe with margin Hysteresis: Derate activates at: T_derate_start = 160 degC Derate fully removed: T_recovery = 145 degC Gap = 15 degC prevents hunting between active/inactive states WITHOUT hysteresis: temperature oscillates either side of T_derate_start causing repeated torque steps -> driveability issue
```

CHAPTER 12

Oil Lifetime Profile

Degradation Mechanisms, Calculation & Testing

12.1 Why EV Oil Lifetime Is Uniquely Challenging

EV powertrain oil faces a unique combination of stresses not encountered in conventional ICE transmissions. Most significantly, many modern EV EDU designs are sealed-for-life — meaning the factory-filled oil must survive the entire vehicle service life of 200,000 km and 15+ years with no oil change. This makes the oil lifetime analysis a direct input to the cooling system design requirement.

Stress Factor	ICE Transmission	EV Powertrain
Electrical stress	None (no live conductors in oil)	HIGH — copper windings bathed in oil
Operating speed	Moderate (<6,000 RPM gears)	VERY HIGH (shaft to 20,000 RPM)
Thermal load	Moderate, cyclic	High, sustained at peak performance
Service interval	60,000–100,000 km	Sealed for life (no drain plug in many designs)
Contamination	Combustion products (soot, fuel)	Copper ions, insulation particles, wear metals
Electrical conductivity	Not a concern	Critical — must remain low throughout life

12.2 The Five Degradation Mechanisms

1 — Thermal Oxidation (Primary)

Thermal oxidation is the dominant oil degradation mode. At elevated temperatures, oil molecules react with dissolved oxygen in a chain reaction that produces acidic byproducts, varnish, and sludge. The Arrhenius relationship means that every 10°C increase in oil temperature approximately doubles the oxidation rate — making sump temperature control a direct life-of-oil requirement.

```
Arrhenius Oxidation Rate: Rate doubles every ~10 degC rise At 100 degC: oil life = L (reference) At 110 degC: oil life = L/2 At 120 degC: oil life = L/4 At 130 degC: oil life = L/8 At 140 degC: oil life = L/16
```

2 — Electrical Degradation (EV-Specific)

Electrical Mechanism	Description	Consequence
Electrostatic discharge erosion	PWM inverter creates shaft voltage → bearing current discharge through oil film	Bearing pitting; metallic particles in oil
Copper ion dissolution	Copper dissolves from winding surfaces into oil; Cu^{2}■ acts as pro-oxidant catalyst	Oxidation rate accelerated 3–10× at 5–10 ppm Cu
Conductivity increase	Degraded oil becomes more electrically conductive	Current leakage paths; insulation risk
Additive electrolysis	Polar additive molecules migrate under electric field	Anti-wear additives depleted at critical surfaces

3 — Mechanical Shear Degradation

High-speed gear mesh and bearing contacts mechanically shear long-chain polymer additives (viscosity index improvers) in the oil, causing permanent viscosity loss. As viscosity drops, the elastohydrodynamic (EHD) oil film thickness at gear teeth and bearing contacts reduces, increasing metal-to-metal contact and wear rate.

```
Oil film thickness (simplified EHD): h_film proportional to (eta x speed) / load If viscosity (eta) drops by 20% due to shear: Film thickness drops by ~20% Wear rate increases exponentially below critical film thickness KRL Shear Test (CEC L-45): EV oil requirement: viscosity loss < 10% after 20 hours
```

4 — Hydrolysis (Water Contamination)

Water enters the oil circuit through condensation during thermal cycling, seal permeation over time, and residual manufacturing moisture. Water reacts with additives causing hydrolysis, rust, emulsification, and — most critically for EVs — forms copper hydroxide with dissolved copper ions, which strongly accelerates oxidation. Water content must be kept below 200 ppm.

5 — Particle Contamination

Particle Size	Effect	Source
>10 µm	Direct bearing damage — three-body abrasive wear	Gear/bearing wear debris, assembly contamination
1–10 µm	Accelerated surface fatigue — micropitting on gear teeth	Seal wear, oxidation products
<1 µm	Catalytic acceleration of oxidation	Insulation particles, very fine wear debris

12.3 Arrhenius Lifetime Calculation

The oil lifetime calculation uses the Arrhenius damage accumulation model. The vehicle mission profile defines how many hours the oil spends at each temperature, and the oil supplier provides oxidation life data at reference temperatures. Cumulative damage is calculated using Miner's rule — the same approach used for structural fatigue life prediction.

Lifetime Damage Calculation

```
Cumulative damage model: D_total = SUM_i (t_i / L_i) t_i = hours spent at temperature T_i (from mission profile simulation) L_i = oil life at constant temperature T_i (from supplier data) Oil FAILS when D_total = 1.0 Example calculation (oil life at 100 degC = 8,000 hours): T= 80 degC: 3000 hrs / 32000 hrs = 0.094 damage T=100 degC: 2000 hrs / 8000 hrs = 0.250 damage T=110 degC: 1000 hrs / 4000 hrs = 0.250 damage T=120 degC: 500 hrs / 2000 hrs = 0.250 damage T=130 degC: 200 hrs / 1000 hrs = 0.200 damage T=140 degC: 50 hrs / 500 hrs = 0.100 damage ----- D_total = 1.144 > 1.0 -> OIL FAILS before end of vehicle life Action required: Reduce sump temperature OR specify higher grade oil
```

12.4 How Oil Lifetime Drives Cooling Design

Oil Lifetime → Cooling Specification Chain

```
Oil lifetime requirement (sealed for life, 200k km) | V Arrhenius calculation: Oil fails if T_sump_max > 130 degC for defined hour distribution | V Thermal model of sump: T_sump = f(P_loss, Q_oil, T_oil_in, T_ambient) | V Oil cooler sizing requirement: Must keep T_sump < 125 degC (5 degC margin) at worst case: - Maximum power demand - 45 degC ambient - Maximum grade climb | V YOUR SPECIFICATIONS: Oil cooler UA value (W/K) Oil flow rate through cooler (L/min) Coolant temperature at cooler inlet (degC)
```

12.5 Oil Selection Criteria Summary

Property	Requirement	Reason
SAE Grade	0W-20 or 0W-16	Low viscosity for efficiency; wide temp range operation
Base oil type	Group III or Group IV (PAO) synthetic	Oxidation resistance; shear stability; low-temp pumpability
Thermal stability	Oxidation resistance to 130°C+	Sealed-for-life requirement
Electrical conductivity	<1000 pS/m throughout life	Prevent current leakage paths near live conductors
Copper corrosion	No Cu corrosion (Cu strip test pass)	Copper dissolution accelerates oxidation
Shear stability	Viscosity loss <10% (KRL test)	Maintain EHD film over vehicle life
Pour point	<–45°C	Cold start pumpability at –40°C ambient
Foam control	Anti-foam additives essential	High-speed churning generates foam — disrupts cooling
Seal compatibility	No swell/shrink of FKM, HNBR	Seal integrity over vehicle life
EP additives	Compatible with copper — no S/P attack	Standard GL-4/5 EP additives attack copper

CHAPTER 13

Stator Winding Temperature

Insulation System, Ageing, Performance Effects & Measurement

13.1 The Slot Structure & Hot Spot

The stator winding hot spot is the single most important temperature in the entire EV powertrain thermal system. It is the temperature at the innermost copper conductor in the deepest slot — the furthest point from any cooling surface. It is never directly measured in production vehicles; the ECU thermal observer estimates it from measurable boundary conditions. All insulation life calculations and derate thresholds refer to this hot spot temperature.

Slot Thermal Resistance Chain

```
Slot thermal resistance chain: [Copper conductor] -- R_enamel --> [Slot varnish/potting] --
R_slot_liner --> [Stator tooth iron] -- R_tooth --> [Stator yoke] -- R_yoke_housing -->
[Housing wall] -- R_housing_coolant --> [Coolant] Thermal conductivities: Copper: 400 W/m.K
(excellent conductor) Epoxy impregnation: 0.2-1.0 W/m.K (moderate – fills voids) Slot liner
(Nomex/Kapton): 0.15-0.25 W/m.K (BOTTLENECK) Silicon steel: 25-40 W/m.K (good conductor)
Aluminium housing: 150-200 W/m.K (good conductor) Slot liner is the dominant resistance: Even
0.2 mm thickness at 0.2 W/m.K creates significant delta_T This is why end-winding oil spray is
so thermally valuable: it bypasses the slot liner resistance entirely
```

13.2 Insulation System Components

Layer	Material	Thickness	Function	Failure Mode
Conductor enamel	Polyimide (PI) or PAI	30–80 µm	Isolate conductors from each other	Turn-to-turn short circuit
Slot liner	Nomex or Kapton film	0.2–0.5 mm	Isolate winding from stator iron (ground)	Winding-to-ground fault
Phase separator	Nomex paper	0.2–0.3 mm	Separate phase conductors in shared slots	Phase-to-phase arc fault
Impregnation varnish	Epoxy or polyester resin	Fills all voids	Mechanical bonding + thermal conductivity + moisture barrier	Cracking → moisture ingress → insulat

13.3 Thermal Ageing — The Montsinger Rule

Insulation does not fail instantly when its class temperature is exceeded. It undergoes cumulative thermal ageing — an irreversible chemical degradation that progressively reduces dielectric strength and mechanical properties. This ageing follows Arrhenius kinetics, described practically by the Montsinger Rule.

```
Montsinger Rule: 'Every 10 degC rise in operating temperature halves insulation
life' Arrhenius form: L(T) = L_ref x exp( Ea/R x (1/T - 1/T_ref) ) L_ref =
insulation life at reference temperature Ea = activation energy (~1.0-1.4 eV for
polyimide) T = operating temperature (Kelvin) Class H example (L_ref = 20,000 hrs
at 180 degC): At 170 degC -> life = 40,000 hrs (2x longer) At 160 degC -> life =
80,000 hrs (4x longer) At 190 degC -> life = 10,000 hrs (2x shorter) At 200 degC
-> life = 5,000 hrs (4x shorter)
```

13.4 Effect of Winding Temperature on Motor Performance

Winding temperature affects motor performance in multiple ways, all of which are thermally coupled — creating a positive feedback loop that must be managed by the derating strategy.

Temperature Effects on Motor Performance

```
Copper resistance temperature dependence: R(T) = R_20 x (1 + alpha x (T - 20)) alpha_copper = 0.00393 per degC At 20 degC: R = R_20 (reference) At 100 degC: R = 1.31 x R_20 (+31%) At 150 degC: R = 1.51 x R_20 (+51%) At 180 degC: R = 1.63 x R_20 (+63%) Consequences: 1. Copper losses INCREASE with temperature P_copper = I^2 x R -> more heat at same current -> thermal runaway risk 2. Torque constant Kt DECREASES slightly (resistance affects dynamic) Same current -> less effective torque production 3. Field weakening speed REDUCES at hot conditions V_dc = E_back + I x R + I x L x omega Higher R -> less voltage headroom -> FW starts at lower speed Customer notices reduced top speed / high speed acceleration when hot
```

13.5 Winding Temperature — Key Threshold Values

Temperature	Significance
20°C	Cold soak reference — calibration baseline; maximum copper resistance accuracy
60°C	Oil typically reaches this temperature before motor on warm-up; VSC may allow full torque
100°C	Copper resistance 31% above cold; efficiency measurably lower; normal operating range
130°C	Typical steady-state at aggressive but normal driving; continuous operation target
155°C	Class F insulation rated limit; derate warning zone in older motor designs
160°C	Typical T_derate_start for Class H motor; controlled torque reduction begins
175°C	Typical T_derate_end; full torque removal; model uncertainty margin before class limit
180°C	Class H continuous rated limit; NOT to be exceeded in normal operation under any condition
200°C	Short-term absolute maximum; insulation survives briefly; life significantly consumed
220°C+	Irreversible insulation damage; slot liner breakdown risk; immediate hardware failure risk

13.6 Measurement Methods in Development

Method	What It Measures	Advantages	Limitations
Embedded thermocouples (Type K/T)	Temperature inside slot and at end winding surface	Direct measurement; fast response	Cannot reach true hot spot; sparse spatial coverage with multiple sensors needed
Resistance Temperature (DC method)	Average copper temperature from winding DC resistance	True average copper temperature; no sensor needed	Average only — not hot spot; not real-time; sto
Infrared thermal camera	End winding surface temperature distribution	Spatial temperature map; identifies hot spots	Surface only — not slot interior; emissivity cali
Fibre optic sensor	Point temperature inside slot with high accuracy	High accuracy; immune to electrical interference	Fragile; expensive; requires special slot prepa

CHAPTER 14

Rotor & Magnet Temperature

Demagnetisation Risk, Cooling Strategies & Thermal Coupling

14.1 Why the Rotor Is the Harder Thermal Problem

Despite often generating less total heat than the stator, the rotor presents a fundamentally harder thermal problem for three reasons: it rotates at up to 20,000 RPM making direct coolant access impossible; the thermal resistance from magnet to coolant is 5–10 times higher than from stator winding to coolant; and the long thermal time constant means rotor temperature can continue rising for minutes after electrical load is removed — the thermal soak phenomenon.

Rotor Thermal Problem

```
Rotor heat path (every arrow is a thermal resistance): [Magnet] -> [Rotor iron] -> [Shaft] -> [Bearing] -> [Housing] -> [Coolant] Comparison: Stator thermal resistance (winding to coolant): ~5-15 K/kW Rotor thermal resistance (magnet to coolant): ~30-80 K/kW Rotor thermal time constant: 100-500 seconds Stator thermal time constant: 30-120 seconds Result: Rotor temperature rises SLOWLY but KEEPS RISING even after electrical load is removed. Post-drive thermal soak is critical consequence.
```

14.2 Magnet Properties vs Temperature

NdFeB (Neodymium Iron Boron) permanent magnets are used in virtually all high-performance EV traction motors. Their magnetic properties — remanence Br and coercivity Hci — both degrade with temperature. Critically, coercivity (the demagnetisation resistance) is far more temperature-sensitive than remanence.

```
Remanence temperature coefficient: Br(T) = Br_20 x (1 + alpha_Br x (T - 20)) alpha_Br (NdFeB) = -0.11 %/degC Coercivity temperature coefficient: Hci(T) = Hci_20 x (1 + alpha_Hci x (T - 20)) alpha_Hci (NdFeB) = -0.55 %/degC (5x more sensitive!) At 140 degC vs 20 degC: Br reduction: 0.11% x 120 = 13.2% -> torque drops ~13% Hci reduction: 0.55% x 120 = 66% -> demag resistance SEVERELY compromised
```

14.3 Demagnetisation — The Irreversible Failure Mode

Demagnetisation is the most feared failure in EV motor design. Unlike insulation degradation which is gradual and potentially detectable, demagnetisation can be sudden and leaves the motor permanently damaged with reduced torque capability that cannot be restored by any repair short of magnet replacement.

Demagnetisation Conditions

```
Two conditions cause irreversible demagnetisation: Condition 1 — High Temperature Alone: Hci decreases -> knee point of B-H curve rises Even without high current, magnet operating point can cross knee point at elevated temperature. Defines: T_demag_limit (material property of magnet grade) Condition 2 — High Current + Elevated Temperature: During field weakening operation, large negative d-axis current creates demagnetising field opposing the magnet. At elevated temperature (low Hci), this field can exceed the knee point -> partial or full demagnetisation. Most critical during: - High speed operation (field weakening region) - Short circuit fault current event - Peak torque demand when rotor already hot Consequence of partial demagnetisation: 5-20% permanent torque reduction Customer experiences: 'car feels permanently less powerful' Cannot be repaired — motor replacement required Significant warranty cost and customer satisfaction impact
```

14.4 Magnet Temperature Thresholds

Temperature	Significance for NdFeB N45SH Grade
20°C	Reference: Br = 1.35 T, Hci = 1600 kA/m — full magnetic properties
80°C	Br reduced ~7%, Hci reduced ~33% — still safe, normal operating range
100°C	Br reduced ~9%, Hci reduced ~44% — elevated; field weakening demag risk rising
110°C	T_derate_start_magnet — begin reducing peak current, especially in field weakening
120°C	Hci reduced ~55% — significant; current limits must be strictly enforced; continuous limit
130°C	T_derate_end_magnet — maximum current sharply reduced; FW capability severely limited
150°C	Absolute maximum N45SH — irreversible demagnetisation risk at any operating current
180°C+	Approaching Curie temperature — complete loss of magnetic ordering; catastrophic failure

14.5 Rotor Cooling Strategies

Strategy	Description	Thermal Effectiveness	Design Challenge
Shaft oil feed	Oil pumped through hollow shaft; centrifugal [illegible]	HIGH — [illegible]	Sealing at rotating [illegible]; uniform distribution; high-sp
End plate oil spray	Oil jets directed at rotor end faces; cools end winding [illegible]	MODERATE — [illegible]	Centrifugal [illegible] at high RPM; cove
Air gap oil mist	Forced oil mist through air gap; direct rotor outer surface contact	HIGH — [illegible]	Major windage loss increase; foam generation; air
Passive conduction via stator	Rotor heat → air gap convection → stator [illegible]	LOW — high air gap resistance	Background path only — not a primary cooling str

14.6 Stator-Rotor Thermal Coupling

Stator and rotor temperatures are not independent — they are thermally coupled through the air gap. This coupling is often underestimated in simplified thermal models and can lead to significant prediction errors, particularly at operating conditions where both stator and rotor are simultaneously hot.

```
Air gap heat transfer: Q_airgap = h_airgap x A_airgap x (T_stator_inner - T_rotor_outer) Air gap convection coefficient h_airgap: Low speed (<1000 RPM): h = 50-100 W/m2.K (conduction dominated) High speed (>10000 RPM): h = 200-500 W/m2.K (Taylor-Couette turbulent) Critical coupling scenario: Stator hot (high torque) AND rotor hot (high speed) -> Air gap can transfer heat FROM rotor TO stator or vice versa -> Rotor thermal model MUST include this coupling term -> Omitting it causes systematic rotor temperature underestimation
```

14.7 Operating Condition vs Thermal Risk Summary

Operating Condition	Primary Thermal Risk	Reason	Cooling Priority
Low speed, high torque (launch, tow)	STATOR WINDING	High I²R; low air gap convection; no [illegible]	Maximum oil flow — end winding spray crit
High speed, light load (motorway cruise)	ROTOR MAGNET	High iron losses at high electrical [illegible]	Shaft oil feed, moderate end winding spray

Operating Condition	Primary Thermal Risk	Reason	Cooling Priority
High speed, high torque (performance)	BOTH simultaneously	Maximum losses everywhere; worst-case combined condition	Maximum both circuits
Regenerative braking	Usually LOW — monitor	Current flows but energy recovered; modest I^2R	VSC minimum maintained; no extra deman
Stationary with torque (hill hold)	STATOR — worst case	Zero speed; zero windage; sustained current; no centrifugal aid	Maximum oil flow; monitor closely; time-lim
Post-drive thermal soak	ROTOR MAGNET	Long rotor time constant; temperature rises after load removed	Post-drive after-run pumping essential

CHAPTER 15

Quick Reference Tables

Key Numbers, Constants & Engineering Rules of Thumb

15.1 Key System Performance Numbers

Parameter	Typical Value	Notes
Motor peak efficiency (SiC inverter)	95–98%	At optimal operating point
Motor continuous efficiency (WLTP avg)	91–94%	Weighted over drive cycle
Inverter efficiency (SiC)	98–99%	Peak; lower at partial load
EDU total efficiency (WLTP)	91–94%	Motor + inverter + gearbox combined
Motor peak power (typical BEV)	150–300 kW	Programme-dependent
Motor continuous power	50–60% of peak	Thermally limited
Peak torque duration	10–30 seconds	From cold; shorter from hot

15.2 Thermal Limits Quick Reference

Component	Continuous Limit	Peak Limit	Hard Absolute Limit
Stator winding (Class F)	155°C	170°C	180°C
Stator winding (Class H)	180°C	200°C	220°C
Rotor magnet (N45SH)	120°C	140°C	150°C
Bearing (standard)	120°C	140°C	160°C
Oil sump	125°C	135°C	145°C
IGBT junction (Si)	150°C	165°C	175°C
SiC MOSFET junction	175°C	185°C	200°C

15.3 Flow Rate Reference

Circuit / Condition	Min Flow	Nominal Flow	Max Flow
Oil — VSC minimum (standstill)	0.5–1.5 L/min	—	—
Oil — continuous operation	2–3 L/min	4–6 L/min	8–10 L/min
Oil — peak performance	6–8 L/min	8–10 L/min	10–12 L/min
Coolant — stator jacket	5–8 L/min	10–15 L/min	18–22 L/min

Circuit / Condition	Min Flow	Nominal Flow	Max Flow
Coolant — inverter cold plate	5–8 L/min	8–12 L/min	15–18 L/min
Coolant — combined EDU	10–12 L/min	15–20 L/min	22–28 L/min

15.4 Key Material Properties

Material	Density (kg/m³)	Cp (J/kg·K)	Thermal Cond. (W/m·K)
Copper (winding)	8,900	385	400
Silicon steel (lamination)	7,600	460	25–40
Aluminium (housing)	2,700	900	150–200
NdFeB magnet	7,500	420	6–9
ATF/EV oil (100°C)	~830	~2100	~0.14
Water-glycol 50/50 (70°C)	~1040	~3700	~0.42
Slot liner (Nomex/Kapton)	~1,400	~1,100	0.15–0.25
Epoxy impregnation	~1,200	~1,000	0.2–1.0

15.5 Copper Resistance Temperature Correction

Winding Temperature	R/R_20 Factor	Copper Loss Multiplier vs Cold
20°C	1.000	1.00× (reference)
60°C	1.157	1.16× — warm motor
100°C	1.314	1.31× — normal operating
130°C	1.431	1.43× — aggressive driving
160°C	1.549	1.55× — derate zone
180°C	1.627	1.63× — class limit

15.6 Magnet Property vs Temperature (NdFeB N45SH)

Temperature	Br (T)	Hci (kA/m)	Relative Demag. Risk
20°C	1.35	1600	Very Low — full properties
60°C	1.31	1380	Low
80°C	1.28	1270	Low-Moderate — normal warm
100°C	1.26	1150	Moderate — FW care needed
120°C	1.23	1020	High — continuous limit

Temperature	Br (T)	Hci (kA/m)	Relative Demag. Risk
140°C	1.21	900	Very High — peak limit
160°C	1.18	780	CRITICAL — avoid under all conditions

15.7 Insulation Class Summary

Class	Max Continuous Temp	Typical Material	EV Application
Class A	105°C	Cotton, paper, natural resins	Not used in EV motors
Class E	120°C	Some synthetic resins	Not used in EV motors
Class B	130°C	Mica, glass fibre, asphalt	Legacy industrial motors
Class F	155°C	Epoxy resins, glass fibre	Early EV designs; budget applications
Class H	180°C	Silicone elastomers, polyimide	Most current production EV motors
Class C	>180°C	Polyimide, PTFE, ceramic	High performance EVs; motorsport
200 / 220	200–220°C	Advanced polyimide systems	Next-gen high-performance EV motors

END OF DOCUMENT

EV Powertrain Cooling & Lubrication — Systems Engineering Reference

Jaguar Land Rover | Powertrain Thermal Systems | May 2026